Samyak Kharat

Portfolio ♦ +91 9022930903 ♦ Pune, MH ♦ LinkedIn

EDUCATION

MIT World Peace University **2024 – 2027**
B.Tech in ECE AI&ML (4th Year) *Pune, MH*

MIT World Peace University **2021 – 2024**
Diploma in ECE Ai&MI *Pune, MH*

WORK EXPERIENCE

Tech Team (MTT), ROBOCON **2023 – 2026**
Robotics Control Developer *Pune, MH*

Lead — Head of the Bot *2026*

- Helped build the step-climbing mechanism and mecanum-drive autonomous navigation system for the team's DD Robocon 2026 robot, a contribution that helped the team finish All India Rank 2.

Mentor — OpenCV Workshop *2025 – 2026*

- Taught OpenCV and computer vision fundamentals through hands-on technical training for 1st- and 2nd-year students as part of the team's internal workshop series.

Senior Engineer *2025*

- Developed the core control system for the jump bot, which the team used to land the season's basketball dunk shot.
- Engineered a real-time auto-aim system using LiDAR, sensor fusion, and odometry, enabling closed-loop self-alignment and dynamic shooting-power adjustment.

Engineer *2023 – 2024*

- Built an autonomous ROS2 robot for ball sorting and storage, developing OpenCV-based navigation and object detection that let it locate and store balls without manual input.
- Bridged hardware and software by integrating STM32 and ESP32 nodes over CAN bus and UART, and built a manually operated robot with an integrated embedded control system, laying the groundwork for the team's later automation efforts.

PROJECTS

DD Robocon (2023 Finalist, 2024 Semi-Finalist, 2025 Dunk Robot, 2026 Finalists)

- Achieved AIR 2 in DD Robocon 2026, AIR 3 in DD Robocon 2024, and AIR 2 in DD Robocon 2023 through competitive robotics design and autonomous systems development.

Vision-Guided Aiming Turret

- Camera-guided turret that detects, tracks, and aims at a moving ball, using ROS2 to drive ODrive motors from real-time OpenCV target detection.

Autonomous Mapping & Navigation Rover

- Differential-drive robot navigating indoor spaces via the Nav2 stack, with SLAM-based mapping sequenced by a custom behavior tree.

STM32 PID Motor Controller

- Closed-loop PID speed and position control for a DC motor on STM32, with gains tuned for stable response under varying load.

ROS2 Hardware Interface for Mecanum-Drive Base

- Custom ros2_control hardware interface bridging ROS2 velocity commands to STM32 firmware over CAN, with closed-loop odometry feedback.

SKILLS

Programming Languages & Tools:
C, C++, Python, Linux, Git, Version Control

Robotics, Simulation & Automation:
ROS2, Gazebo, Rviz, ros2_control, Nav2, SLAM, Motion Planning, Autonomous Navigation, Sensor Fusion, Docker, Ansible, GitHub Actions, CI/CD, MuJoCo

Computer Vision:
OpenCV, Object Detection, Image Processing, Real-Time Vision Systems

Embedded Systems:
STM32, TM4C, ESP32, Embedded C, Microcontroller Programming, CAN Bus, UART, PWM, Real-Time Control Systems